Statistics Guide — algorithms and methods behind Tab 3

This document explains **how** each column in the Tab 3 "Generate Statistics" CSV is computed — the specific algorithm, library, and formula behind it — not just what the number means. For a plain-English, biology-oriented explanation of each column instead, see GUIDE.md, Section 10. This document is the companion reference for anyone extending `_statistics.py`, auditing a result, or writing up the method (e.g. a Methods section).

All of this lives in one module, `napari_zf_microglia_ai/_statistics.py`, entered through a single function, `compute_stats()`.

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1. Pipeline architecture

`compute_stats(labels, scale_zyx, image=None, region_lines=None, region_names=None, backend_config=None)` runs in three phases, each chosen to keep the expensive parts vectorised or parallelised rather than looping over labels one at a time in pure Python:

Phase	What	How
1	Batch regionprops (volume, centroid, bbox, inertia tensor, solidity, extent) for **every** label in one call	cuCIM (GPU) if available, else scikit-image (CPU)
2a	Per-label marching-cubes surface area + skeleton/branch analysis	`ThreadPoolExecutor`, one worker per label, cropped to that label's bounding box
2b	Per-label intensity statistics *(only if an Image layer was supplied)*	Same thread-pool pattern as 2a
3	Assemble the DataFrame, then compute statistics that need *all* labels at once (nearest-neighbour, brain region), then generate descriptions	Vectorised NumPy/SciPy, then one description call per row

Why cropping to the bounding box matters (2a/2b): marching cubes and skeletonization are run on a small 3D crop (`labels[z0:z1, y0:y1, x0:x1] == lbl`), not the full volume — this is what makes per-label operations on a multi-megavoxel image tractable at all. The crop bounds come directly from Phase 1's regionprops `bbox` output, so Phase 1 must run first.

GPU backend detection (`_detect_stats_backend()`): tries importing `cupy` + `cucim.skimage.measure`, then runs a 4×4×4 smoke-test `regionprops_table` call to confirm the CUDA JIT actually compiles and runs (not just that the import succeeded) before committing to the GPU path for the whole run. Falls back to CPU

(`skimage.measure.regionprops_table`) on any failure — including cuCIM being unavailable on Windows, since it has no native Windows wheels (see README/GUIDE's Windows install notes).

Thread count: `_N_THREADS = max(1, os.cpu_count() // 2)` — half the logical cores, leaving headroom for the GPU-bound phase and the main thread rather than saturating every core.

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2. Phase 1 — Batch regionprops

`_batch_regionprops(labels)` calls `regionprops_table` (cuCIM or skimage — same API, same property names) once, requesting:

```
_RPROPS = [
    "label", "area", "centroid", "bbox",
    "inertia_tensor", "inertia_tensor_eigvals",
    "axis_major_length", "axis_minor_length",
    "solidity", "extent",
]
```

This single call is where most of the raw geometry comes from:

Column(s)	Source property	Notes
`volume_vox`	`area`	scikit-image calls the 3D voxel count "area" for historical (2D-first) API reasons
`centroid_z/y/x_vox`, `centroid_z/y/x_um`	`centroid`	voxel centroid × `scale_zyx` for the µm columns
`bbox_z0/y0/x0_vox` (start, inclusive), `bbox_z1/y1/x1_vox` (end, exclusive)	`bbox`	standard NumPy half-open slicing convention — `labels[z0:z1, y0:y1, x0:x1]` is exactly the label's bounding crop
`bbox_dz/dy/dx_um`	`bbox` (end − start) × `scale_zyx`	physical size of the bounding box, not the label's own extent (a diagonal or branched cell can have a much larger

Column(s)	Source property	Notes
		bbox than its actual volume would suggest — see `extent` below)
`solidity`	`solidity` directly	`volume / convex_hull_volume` — the library computes the 3D convex hull internally and divides
`extent`	`extent` directly	`volume / bounding_box_volume`
`axis_major_length`, `axis_minor_length`	used below	**pixel-space** ellipsoid-fit lengths, not yet in µm — see §3 for why these need care under anisotropic voxels

`volume_um3 = volume_vox × sz × sy × sx` (`vox_vol` = the physical volume of one voxel, computed once from `scale_zyx`).

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3. Derived shape metrics from the inertia tensor

This is the part of the pipeline most worth reading carefully if you're citing these numbers, because it involves a real approximation under anisotropic voxel scaling (Z = 1.0 µm/vox vs. Y = X = 0.174 µm/vox in this project's typical data).

3.1 `principal_axis_dir` — which axis is the cell elongated along

The 3×3 inertia tensor for every label (from Phase 1) is stacked into an `(N, 3, 3)` array and eigendecomposed in one batched call, `np.linalg.eigh(IT)`. `eigh` returns eigenvalues in **ascending** order — physically, the eigenvector paired with the **smallest** eigenvalue of the

inertia tensor is the object's axis of elongation (mass concentrated close to that axis contributes little to inertia about it; mass spread far from an axis contributes a lot). So:

```
longest_vecs = eigvecs[:, :, 0]   # smallest-eigenvalue
        eigenvector = major axis direction
axis_dirs = ["Z","Y","X"][argmax(|component|)]  # per label,
        whichever axis the eigenvector points closest to
```

`principal_axis_dir` is therefore the axis (Z, Y, or X) that the true 3D major axis most nearly aligns with — not a continuous angle, just the dominant of the three.

3.2 `axis1_um`, `axis2_um`, `axis3_um` — the anisotropy caveat

`axis_major_length`/`axis_minor_length` from regionprops are computed in **voxel units**, from an ellipsoid fit that implicitly assumes isotropic voxels. Converting to µm correctly requires knowing *which physical axis* the length actually runs along — a length "along Z" and a length "along X" need different scale factors (1.0 vs. 0.174 µm/vox here) to become physically correct.

The code handles this only for the *major* axis, by construction:

```
axis1_scales = [axis_scale_map[d] for d in axis_dirs]
        # pick sz, sy, or sx per-label, matching
        principal_axis_dir
a1_um = axis_major_length * axis1_scales
a3_um = axis_minor_length * scale_mean                    #
        scale_mean = (sz+sy+sx)/3 — an approximation
a2_um = (a1_um + a3_um) / 2.0                              # NOT
        independently measured — see below
```

Three things worth knowing if you rely on these numbers:

- **`axis1_um` is scaled correctly** (per-label, using whichever physical axis it actually points along).
- **`axis3_um` uses the mean of the three axis scales**, not the true axis it points along, because regionprops' minor-axis direction isn't tracked per-label the way the major axis is here. This is an approximation, more accurate the closer `sz`/`sy`/`sx` are to each other (it is *not* a negligible approximation for this project's typical ~5.7:1 Z:XY anisotropy on an axis pointing mostly along Z or mostly along XY).
- **`axis2_um` is not measured at all** — it's simply `(axis1_um + axis3_um) / 2`, a placeholder for the true middle principal axis (which would require a second eigenvector-aligned length

measurement that regionprops doesn't provide directly). Treat `axis2_um` as illustrative, not a real geometric measurement.

3.3 `elongation`, `eq_diam_um`

- `elongation = axis1_um / axis3_um` (guarded against divide-by-zero with a `1e-10` floor). 1.0 = sphere; higher = more elongated.
- `eq_diam_um = (6·volume_um3 / π)^(1/3)` — the diameter of a sphere with the same volume as the label. Shape-independent; two cells with identical volume have the same `eq_diam_um` regardless of how elongated or branched either one is.

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4. Phase 2a — Surface area (marching cubes) and sphericity

`_surface_area(binary, scale_zyx)`:

1. The label's cropped binary mask is **padded by 1 voxel on every side** (`np.pad(binary, 1)`) before meshing. This is necessary because marching cubes needs a zero-boundary to close the mesh — without padding, a label touching its own crop's edge would produce an open (non-manifold) surface and an undercounted area.
2. `skimage.measure.marching_cubes(padded, level=0.5, spacing=scale_zyx)` extracts an isosurface at the 0.5 threshold (the boundary between labeled and unlabeled voxels), directly in physical units via the `spacing` parameter (so no separate unit conversion is needed afterward).
3. `skimage.measure.mesh_surface_area(verts, faces)` sums the area of every triangle in the resulting mesh.
4. Returns `0.0` on any failure (e.g. a 1-voxel object with no meaningful surface) rather than raising, so one bad label doesn't abort the whole run.

Sphericity (computed in the main assembly loop, not inside `_surface_area`):

```
sphericity = π^(1/3) · (6V)^(2/3) / A          (V = volume_um3, A = surface_area_um2)
```

This is the classical Wadell (1935) sphericity — the surface area a perfect sphere of the same volume *would* have, divided by the actual surface area. It's clamped to a maximum of 1.0 (mesh discretisation

noise can occasionally push the raw ratio fractionally above 1.0 for near-spherical labels; a physical sphericity cannot exceed 1.0 by definition).

`surface_to_volume_ratio = surface_area_um2 / volume_um3` — a simpler, non-normalised alternative to sphericity; unlike sphericity it keeps growing without bound as branching increases, rather than saturating.

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5. Phase 2a — Skeleton and branching statistics

`_skeleton_stats(binary, scale_zyx)`, using `skimage.morphology.skeletonize` + the skan package:

1. `skeletonize(binary)` — thins the 3D binary mask to a topological skeleton (1-voxel-wide medial curve), preserving connectivity and branch structure.
2. `skan.Skeleton(skeleton, spacing=scale_zyx, source_image=binary)` builds a graph representation, with edges already in physical units via `spacing`.
3. `skan.summarize(sk, separator="-")` produces one row per skeleton **branch** (an edge between two graph nodes — a branch point or an endpoint), with columns including `branch-type`, `euclidean-distance`, and `branch-distance` (path length along the branch).

From that branch table:

Column	Computation
`n_branches`	`len(branch_table)` — one row per branch
`n_endpoints`	count of rows where `branch-type == 1` (skan's code for a branch ending in a free tip, as opposed to a branch connecting two junctions)
`mean_branch_len_um`	mean of `euclidean-distance` (straight-line branch endpoint-to-endpoint distance, **not** path length) across all branches
`max_branch_len_um`	

Column	Computation
	max of the same `euclidean-distance` column
`branch_tortuosity`	mean of `branch-distance / euclidean-distance` (path length ÷ straight-line distance) over branches with nonzero `euclidean-distance`; 1.0 = perfectly straight
`branch_density`	`n_branches / volume_um3 × 10⁶` (branches per million μm³, i.e. per ~100×100×100 μm cube — the ×10⁶ scaling exists purely to keep the numbers in a readable range)
`endpoint_density`	same normalisation, for `n_endpoints`
`process_complexity`	`n_endpoints × mean_branch_len_um / volume_um3` — a single combined index: more endpoints *and* longer branches *and* smaller cell body all push this up

A real gotcha, already found and fixed in this project once (see the branch-radius calibration work): `skan`'s `source_image=` parameter does **not** feed `summarize()`'s per-branch intensity/"mean-pixel-value" columns the way its name suggests — that requires baking the values into the skeleton array itself before constructing `skan.Skeleton`, not passing them as `source_image`. This module doesn't currently use any `source_image`-derived column (only geometry columns), so it isn't affected, but it's a real API trap worth knowing if this module is ever extended to pull intensity-along-skeleton statistics.

Failure mode: if `skan` isn't installed, or skeletonization produces an empty skeleton (can happen for a 1-2 voxel label), `_skeleton_stats` returns `(0, 0, 0.0, 0.0, 1.0)` rather than raising — all branch-derived columns read as zero/neutral for that label instead of aborting the batch.

6. Phase 2b — Intensity statistics (optional)

Only computed when `compute_stats()` is called with an `image` array (Tab 3's Image-layer selector). `_intensity_stats_worker` crops both the label mask and the source image to the label's bounding box, then:

```
mean_intensity        = mean(image[mask])
integrated_intensity  = sum(image[mask])
intensity_cv          = std(image[mask]) / mean(image[mask])
(0.0 if mean == 0)
```

`intensity_cv` (coefficient of variation) is scale-independent — it measures relative spread of brightness within the cell, not absolute brightness, so it's comparable across cells or samples with different overall exposure/gain.

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7. Post-assembly — Spatial statistics

`_spatial_stats(centroids_zyx_um, label_ids)` runs once on **all** centroids together (not per-label), using `scipy.spatial.cKDTree` for efficient nearest-neighbour queries — an O(N log N) k-d tree build rather than an O(N^2) all-pairs distance matrix.

7.1 Nearest neighbours

```
tree = cKDTree(centroids)
dists, idxs = tree.query(centroids, k=3)   # self + 1st NN + 2nd
        NN
```

Column 0 of the result is always the point itself (distance 0), so the 1st nearest neighbour is column 1 and the 2nd is column 2. `nearest_neighbor_label`/`nearest_neighbor_2_label` map the neighbour's *tree index* back to its actual label ID (tree order and label order aren't guaranteed to match). With exactly 2 cells total, `k` is clamped to 2 and the "2nd nearest neighbour" columns fall back to duplicating the 1st (there is no second neighbour to find).

7.2 Clark-Evans 3D index (`nearest_neighbor_ratio`)

The classical Clark & Evans (1954) index compares an observed mean nearest-neighbour distance to the value expected under **complete spatial randomness** (a homogeneous 3D Poisson point process) at the

same point density. This module computes a **per-cell** version — each cell's own NND divided by the population's expected NND — rather than the traditional single population-level summary statistic.

Derivation of the expected 3D nearest-neighbour distance, since this is worth having written down once: for a 3D Poisson process at intensity (density) ρ, the probability that no other point lies within radius *r* of a given point is the Poisson void probability for a sphere of volume $(4/3)\pi r^3$:

$$P(\text{NND} > r) = \exp(-\rho \cdot (4/3)\pi r^3)$$

The expectation of a nonnegative random variable is $E[X] = \int_0^\infty P(X > r)\, dr$, so:

$$E[\text{NND}] = \int_0^\infty \exp(-\rho \cdot (4/3)\pi \cdot r^3)\, dr$$

Substituting $u = r^3$ and using the standard Gamma-function integral $\int_0^\infty \exp(-a \cdot r^3)\, dr = (1/3) \cdot \Gamma(1/3) \cdot a^{(-1/3)}$, with $a = \rho(4/3)\pi$, and the identity $\Gamma(4/3) = (1/3)\Gamma(1/3)$:

$$E[\text{NND}] = \Gamma(4/3) \cdot (3 / (4\pi \cdot \rho))^{(1/3)}$$

— exactly what the code computes: `math.gamma(4/3) * (3.0 / (4.0*math.pi*density))**(1.0/3.0)`.

`nearest_neighbor_ratio = observed_NND / E[NND]`. Values < **1** indicate the cell is closer to its neighbour than random chance would predict (local clustering); > **1** indicates more spread out than random (regularity/dispersion).

A real approximation to know about: `density = N / bbox_volume`, where `bbox_volume` is the axis-aligned bounding box of the *centroid point cloud itself* (`max - min` per axis, multiplied together) — not the true tissue/brain volume the cells actually live in. If cells are sparse near the edges of their own bounding box (very likely for a roughly ellipsoidal brain region), this systematically **overestimates** density and therefore **underestimates** the expected NND, biasing `nearest_neighbor_ratio` slightly upward (toward "more regular than it really is") near the population edges. Fine as a same-sample relative measure; treat with more caution when comparing the raw ratio across samples with different cell-count-derived bounding boxes.

7.3 `local_density_100um`

`tree.query_ball_point(centroids, r=100.0, return_length=True) - 1` — count of other centroids within a 100 µm radius sphere of each cell (the `-1` removes the point counting itself). A direct local crowding count, independent of the Clark-Evans model assumptions above.

7.4 `depth_normalized`

`clip(centroid_z_um / max(centroid_z_um), 0, 1)` across the current label set — 0 = the shallowest (lowest Z) cell in this result set, 1 = the deepest. Recomputed per-run, so it is **not** comparable across different fish/samples unless they share the same Z range by construction — it's a within-sample relative depth, not an absolute one.

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8. Post-assembly — Brain region assignment (optional)

Only computed when Tab 3 is given a Shapes layer of boundary lines/polylines (drawn by the user, one curve per boundary between two anatomical regions) and a list of region names.

8.1 Geometric primitive: point-to-polyline projection

`_polyline_side_and_dist(cy, cx, pts)` finds, for a 2D point and a multi-segment polyline, which *segment* is nearest and which *side* of that segment the point falls on:

1. For every consecutive vertex pair `(a, b)` in the polyline, project the point onto the **infinite line** through `a→b`, then clamp the projection parameter `t` to `[0, 1]` — this restricts the projection to the actual segment, not the infinite line (standard point-to-segment-distance technique).
2. Track the minimum distance across all segments — that segment is the "nearest segment."
3. At that nearest segment, compute the 2D cross product `dx_seg·(cy − ay) − dy_seg·(cx − ax)`. Its **sign** tells you which side of the segment's direction the point is on: this project's fish are oriented head (small X) → tail (large X), with boundary curves drawn top→bottom (increasing Y) by convention (see GUIDE.md §7a), so a

negative cross product means the point is posterior to (right of) that boundary.

8.2 Region index from multiple boundaries

`_assign_brain_regions` sorts all supplied boundary curves by their **mean X coordinate** (ascending = most anterior first) — this makes boundary order well-defined regardless of the order the user drew them in. For each cell centroid, it counts how many boundaries the cell is posterior to:

```
idx = 0
for boundary in boundaries_sorted_anterior_to_posterior:
    if point_is_posterior_to(boundary):
        idx += 1
region = region_names[idx]
```

With *k* boundary curves this naturally produces *k+1* regions (e.g. 1 boundary → anterior/posterior; 2 boundaries → 3 regions). `region_boundary_dist_um` is simply the minimum distance (in µm) from the cell's centroid to *any* boundary curve — cells near this value are close to a region edge and might reasonably be considered ambiguous/borderline.

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9. Morphotype classification

`_classify_morphotype(sphericity, solidity, elongation, n_branches, surface_to_volume_ratio)` — a hand-tuned rule-based decision list, evaluated in priority order (first matching rule wins):

```
if elongation > 3.5 and n_branches <= 3:
        return "Rod-shaped"
if sphericity > 0.70 and solidity > 0.80 and n_branches <=
        2:      return "Amoeboid"
if n_branches >= 6 and sav > 2.0 and sphericity <
        0.55:           return "Ramified"
if n_branches >= 4 and sphericity <
        0.65:                           return "Intermediate-
        ramified"
else:
        return "Intermediate"
```

This is **not** a fitted/learned classifier — it's a fixed set of thresholds chosen to match the standard microglia morphological categories used in the field (ramified/resting vs. amoeboid/activated, plus rod-shaped

and intermediate states), evaluated directly on this module's own geometric outputs. Because it's rule-based and fully deterministic, it's reproducible and auditable, but it hasn't been validated against an independent ground-truth morphotype labeling — treat it as a descriptive heuristic, not a diagnostic classification, unless/until it's been checked against expert-annotated examples.

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10. Natural-language description generation

The `description` column is produced by one of four interchangeable backends, selected via `backend_config["backend"]`:

Backend	Method	Requires
`rule` (default)	`_rule_based_description()` — a template that stitches together shape/surface/branch phrases from fixed thresholds on the already-computed columns (sphericity, solidity, elongation, branch/endpoint counts, morphotype)	nothing — fully offline, deterministic
`ollama`	POSTs a formatted prompt to a local Ollama server's `/api/generate` endpoint	a running local Ollama install, no API key
`openai`	POSTs to `{api_url}/v1/chat/completions` (OpenAI-compatible chat endpoint)	an API key
`claude`	POSTs to `https://api.anthropic.com/v1/messages`	an API key

The three API backends all format the same shared prompt template (`_STATS_PROMPT`) with that row's already-computed statistics (volume, elongation, dominant axis, sphericity, solidity, branch/endpoint counts, mean branch length, morphotype, centroid) and ask for **one sentence, max 40 words**. Every backend call is wrapped in a `try/except` that returns an inline `"[Backend error: ...]"` string on failure (timeout, bad key, unreachable server) rather than raising — one row's description failing doesn't abort the whole statistics run.

The `rule`-based backend is the only one exercised by anything else in the pipeline (it has no external dependency), and is what runs by default with no configuration.

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11. Full column → algorithm cross-reference

Column	Section
`label`, `volume_vox`, `volume_um3`, `centroid_*`, `bbox_*`	§2
`solidity`, `extent`	§2
`principal_axis_dir`	§3.1
`axis1_um`, `axis2_um`, `axis3_um`	§3.2
`elongation`, `eq_diam_um`	§3.3
`surface_area_um2`, `sphericity`, `surface_to_volume_ratio`	§4
`n_branches`, `n_endpoints`, `mean_branch_len_um`, `max_branch_len_um`, `branch_tortuosity`, `branch_density`, `endpoint_density`, `process_complexity`	§5
`mean_intensity`, `integrated_intensity`, `intensity_cv`	§6
`nearest_neighbor_label`, `nearest_neighbor_dist_um`, `nearest_neighbor_2_label`, `nearest_neighbor_2_dist_um`	§7.1
`nearest_neighbor_ratio`	§7.2
`local_density_100um`	§7.3
`depth_normalized`	§7.4
`brain_region`, `region_boundary_dist_um`	§8
`morphotype`	§9
`description`	§10

For the plain-English meaning of every column (rather than the algorithm behind it), see GUIDE.md §10.

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12. References

- Wadell, H. (1935). *Volume, Shape, and Roundness of Quartz Particles.* Journal of Geology, 43(3) — the sphericity formula used in §4.
- Lorensen, W. E., & Cline, H. E. (1987). *Marching Cubes: A High Resolution 3D Surface Construction Algorithm.* ACM SIGGRAPH Computer Graphics, 21(4) — the surface-meshing algorithm behind `surface_area_um2` (§4).
- Clark, P. J., & Evans, F. C. (1954). *Distance to Nearest Neighbor as a Measure of Spatial Relationships in Populations.* Ecology, 35(4) — the spatial-randomness index generalised to 3D in §7.2.
- Nunez-Iglesias, J., et al. — `skan`: skeleton analysis in Python, used throughout §5.
- van der Walt, S., et al. (2014). *scikit-image: image processing in Python.* PeerJ 2:e453 — `regionprops_table`, `marching_cubes`, `skeletonize` (§2, §4, §5).
- Virtanen, P., et al. (2020). *SciPy 1.0.* Nature Methods — `cKDTree` (§7).